Multiple Regression

Introduction

In multilinear regression, we assume that the average y-values are a (affine) linear function of p explanatory variables $x_1, \ldots, x_p$:

$$\mu_y = \beta_0 + \beta_1 x_1 + \ldots + \beta_p x_p.$$

We can use the `lm()` command in R to estimate the coefficients in the model.

Example: 2018 High Bridge half-marathon

```r
results <- read.csv("http://people.hsc.edu/faculty-staff/bl
head(results)
```

```
##   place bib gender age state    time minutes
## 1     1  66      M  35    VA 1:10:28   70.47
## 2     2  87      M  29    VA 1:18:08   78.13
## 3     3 112      F  32    VA 1:25:47   85.78
## 4     4 116      M  32    VA 1:27:02   87.03
## 5     5  32      M  38    VA 1:27:14   87.23
## 6     6 115      F  31    VA 1:28:15   88.25
```

Making a linear model

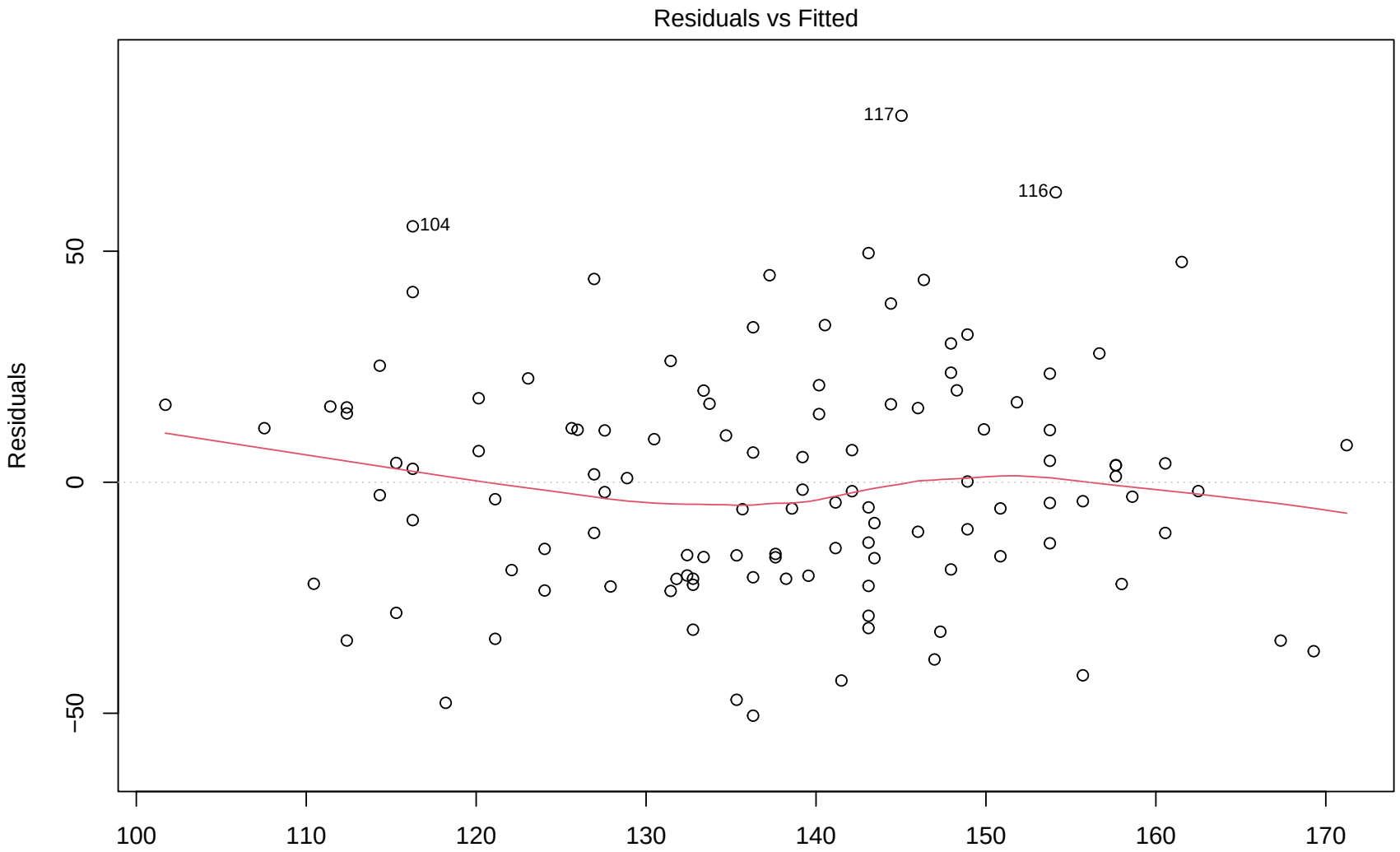
```r
myLM <- lm(minutes ~ age + gender, data = results)
summary(myLM)
```

```
##
## Call:
## lm(formula = minutes ~ age + gender, data = results)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -50.518 -18.871  -1.939  16.194  79.340
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 105.2479     8.6224  12.206  < 2e-16 ***
## age           0.9703     0.1921   5.051 1.68e-06 ***
## genderM     -21.0004     4.7174  -4.452 2.00e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1
```

What does this do?

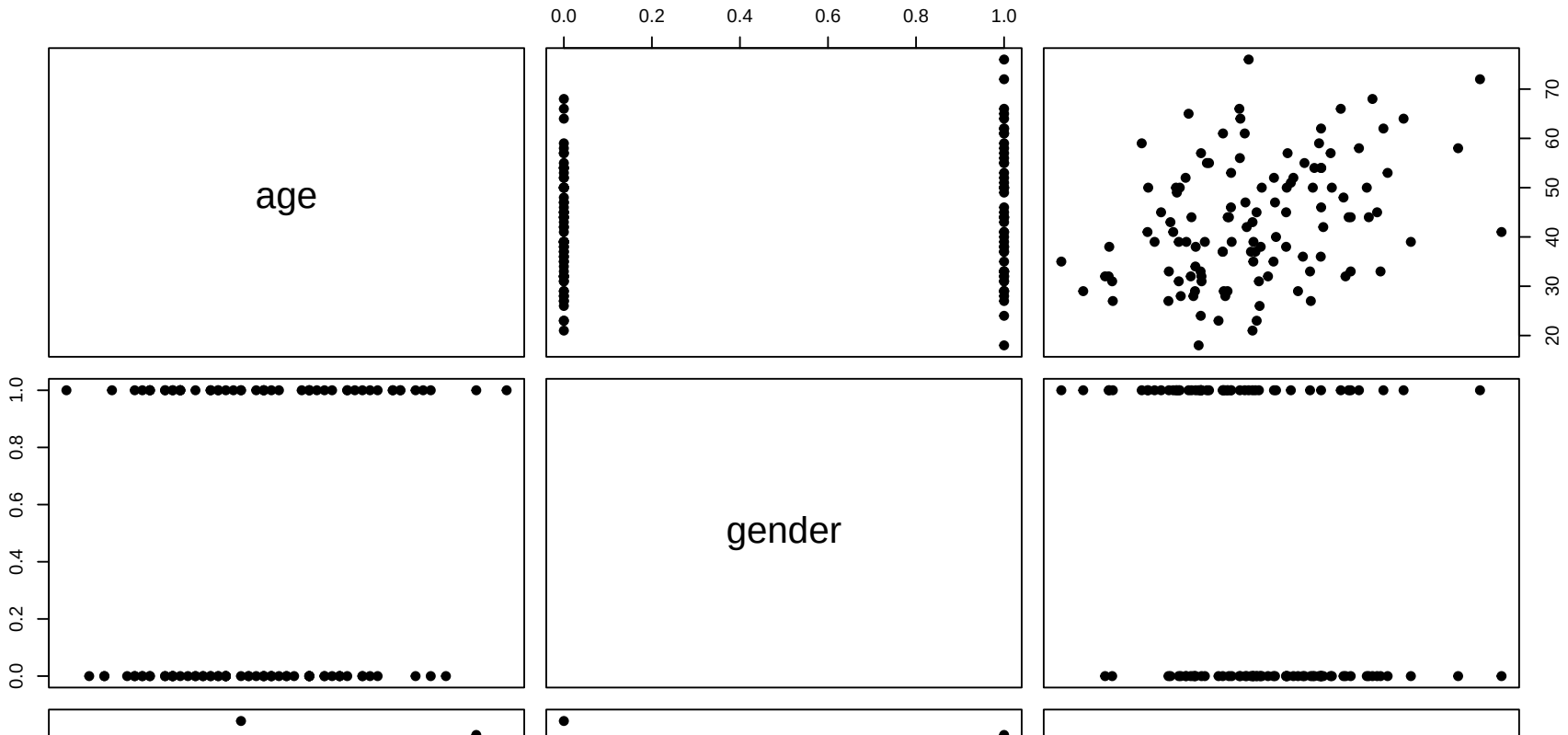
You can plot the model, and you get several images that let you check some of the assumptions of least squares regression.

```
plot(myLM)
```



Comparing pairs of variables

```r
library(dplyr)
#numeric_cols <- sapply(results, is.numeric)
#pairs(results[, c(3,4)])
results |>
  select(age, gender, minutes) |>
  mutate(gender = as.integer(gender == "M")) |>
  pairs(pch = 19)
```



Example: Course evaluations

What factors explain differences in teaching evaluation scores? Here is data from 463 courses at the University of Texas - Austin that can be found in the R library `moderndive` (from the online textbook ModernDive)

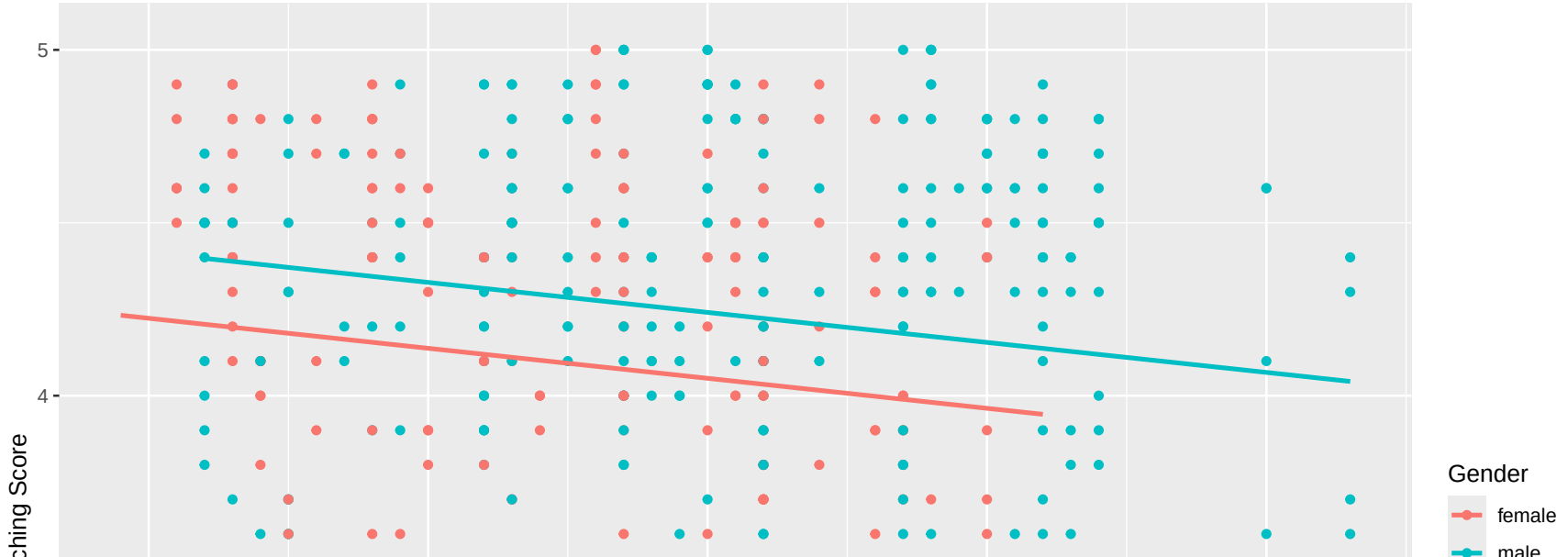
```r
library(moderndive)
evals_data <- select(evals, ID, score, age, gender)
head(evals_data)
```

```
## # A tibble: 6 x 4
##      ID score   age gender
##   <int> <dbl> <int> <fct>
## 1     1   4.7    36 female
## 2     2   4.1    36 female
## 3     3   3.9    36 female
## 4     4   4.8    36 female
## 5     5   4.6    59 male
## 6     6   4.3    59 male
```

A simple linear model

```r
simple_lm <- lm(score ~ age + gender, data = evals_data)
simple_lm
```

```
##
## Call:
## lm(formula = score ~ age + gender, data = evals_data)
##
## Coefficients:
## (Intercept)          age   gendermale
##    4.484116    -0.008678     0.190571
```



Linear models with interaction

A better linear model might have different slopes for men versus women. You can get models like that by adding an iteraction term:

$$\mu_y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_{12} x_1 x_2.$$

Interaction model

```
interaction_lm <- lm(score ~ age * gender, data = evals_dat
interaction_lm
```

```
##
## Call:
## lm(formula = score ~ age * gender, data = evals_data)
##
## Coefficients:
##    (Intercept)              age       gendermale  age:genc
##        4.88299         -0.01752         -0.44604         (
```

Example: Babies birthweights

```
baby_data <- read.csv('http://people.hsc.edu/faculty-staff/
dim(baby_data)
```

```
## [1] 1236    8
```

The `baby_data` data frame contains data on baby birthweights. The variables are:

- bwt - birth weight (in ounces)
- gestation - length of the pregnancy (in days)
- parity - 1 if baby was first born, 0 otherwise
- age - mother's age (in years)
- height - mother's height (in inches)
- weight - mother's weight (in lbs.)
- smoke - 1 if the mother is a smoker, 0 otherwise

Cleaning up the data

There are a lot of cells with NA (not available) entries, and these could mess up our analysis below. The `na.omit()` command is a fast way to remove these.

```
baby_data <- na.omit(baby_data)
dim(baby_data)
```

```
## [1] 1174    8
```

Whenever you omit data, you should make sure that you aren't omitting a large percentage of the sample, and you might also want to check the way that the data was collected to make sure that individuals with missing data are not systematically different from other individuals in the sample. In this example, we are omitting 62 rows of data (out of 1236). That's only about 5% of the data, so we probably aren't affecting our results too much.

The linear model

```r
babies_lm <- lm(bwt ~ gestation + parity + age + height + w
babies_lm
```

```
##
## Call:
## lm(formula = bwt ~ gestation + parity + age + height + w
##     smoke, data = baby_data)
##
## Coefficients:
## (Intercept)    gestation       parity          age
##   -80.41085      0.44398     -3.32720     -0.00895
##       smoke
##    -8.40073
```

Not every factor is significant

```r
summary(babies_lm)
```

```
##
## Call:
## lm(formula = bwt ~ gestation + parity + age + height + w
##     smoke, data = baby_data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -57.613 -10.189  -0.135   9.683  51.713
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -80.41085   14.34657  -5.605 2.60e-08 ***
## gestation     0.44398    0.02910  15.258  < 2e-16 ***
## parity       -3.32720    1.12895  -2.947  0.00327 **
## age          -0.00895    0.08582  -0.104  0.91696
## height        1.15402    0.20502   5.629 2.27e-08 ***
```